FATE MONTHLY REPORT
Doc.No : FATE-001 Version : 2.0 Date :

FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

Performance assessment

Annex

August 2024

PO102958/ESO/20/95952/FLAB

Contents

1 NIGHT1: Contingency tables

Tables in section 1 are included in the range: 1-22.

wind speed ms^{-1}		Observations ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	4772	1852	163
	4.2 < ws < 7.6	2396	3386	1467
	ws > 7.6	231	1512	7702
PC=67.5%; EBD=1.7%; POD1=64.5%; POD2=50.2%; POD3=82.5%				

Table 1: night1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}		Observations ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	6951	465	0
	4.2 < ws < 7.6	448	5972	352
	ws > 7.6	0	313	8980
PC=93.3%; EBD=0.0%; POD1=93.9%; POD2=88.5%; POD3=96.2%				

Table 2: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed ms^{-1}		Observations ws < 12.0	12.0 < ws < 18.0	ws > 18.0
Model data	ws < 12.0	19921	234	0
	12.0 < ws < 18.0	146	2576	88
	ws > 18.0	0	75	441
PC=97.7%; EBD=0.0%; POD1=99.3%; POD2=89.3%; POD3=83.4%				

Table 3: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0) ms^{-1}

wind direction degree		Observations North	East	South	West
Model data	North	11843	663	55	132
	East	661	1165	538	4
	South	90	262	3170	263
	West	375	79	109	223
PC=83.5%; EBD=1.2%; POD1=91.3%; POD2=53.7%; POD3=81.9%; POD4=35.9%					

Table 4: night1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		Observations North	East	South	West
Model data	North	12811	114	0	20
	East	145	1979	41	0
	South	0	76	3822	21
	West	13	0	9	581
PC=97.8%; EBD=0.0%; POD1=98.8%; POD2=91.2%; POD3=98.7%; POD4=93.4%					

Table 5: night1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		Observations rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	4205	1330	37
	7.1 < rh < 15.3	3267	5088	1112
	rh > 15.3	601	1937	5667
PC=64.4%; EBD=2.7%; POD1=52.1%; POD2=60.9%; POD3=83.1%				

Table 6: night1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		Observations rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	7544	315	0
	7.1 < rh < 15.3	529	7841	190
	rh > 15.3	0	199	6626
PC=94.7%; EBD=0.0%; POD1=93.4%; POD2=93.8%; POD3=97.2%				

Table 7: night1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		Observations rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	22675	49	0
	50.0 < rh < 70.0	21	343	6
	rh > 70.0	0	4	146
PC=99.7%; EBD=0.0%; POD1=99.9%; POD2=86.6%; POD3=96.1%				

Table 8: night1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor		Observations		
mm		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
	pwv < 1.9	6256	0	1148
Model data	1.9 < pwv < 3.5	0	0	0
	pwv > 3.5	789	0	14311
PC=91.4%; EBD=8.6%; POD1=88.8%; POD2=NaN%; POD3=92.6%				

Table 9: night1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor		Observations		
mm		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
	pwv < 1.9	6927	137	0
Model data	1.9 < pwv < 3.5	118	6387	86
	pwv > 3.5	0	80	8769
PC=98.1%; EBD=0.0%; POD1=98.3%; POD2=96.7%; POD3=99.0%				

Table 10: night1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
	see < 0.6	3241	2306	872
Model data	0.6 < see < 0.8	1780	2453	2479
	see > 0.8	365	1230	3095
PC=49.3%; EBD=6.9%; POD1=60.2%; POD2=41.0%; POD3=48.0%				

Table 11: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
	see < 0.6	5376	830	6
Model data	0.6 < see < 0.8	682	4692	630
	see > 0.8	2	600	6193
PC=85.5%; EBD=0.0%; POD1=88.7%; POD2=76.6%; POD3=90.7%				

Table 12: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
	see < 0.6	5117	0	568
Model data	0.6 < see < 0.8	0	5808	568
	see > 0.8	269	181	5878
PC=94.3%; EBD=4.7%; POD1=95.0%; POD2=97.0%; POD3=83.8%				

Table 13: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
	see < 0.6	6059	4	3
Model data	0.6 < see < 0.8	0	6107	3
	see > 0.8	1	11	6826
PC=99.9%; EBD=0.0%; POD1=100.0%; POD2=99.8%; POD3=99.9%				

Table 14: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.24

wavefront coherence time		Observations		
ms		tau < 3.4	3.4 < tau < 5.8	tau > 5.8
Model data	tau < 3.4	3676	1323	136
	3.4 < tau < 5.8	1001	2221	842
	tau > 5.8	108	1682	5044
PC=68.2%; EBD=1.5%; POD1=76.8%; POD2=42.5%; POD3=83.8%				

Table 15: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		tau < 3.4	3.4 < tau < 5.8	tau > 5.8
Model data	tau < 3.4	4491	303	0
	3.4 < tau < 5.8	294	4532	334
	tau > 5.8	0	391	5688
PC=91.8%; EBD=0.0%; POD1=93.9%; POD2=86.7%; POD3=94.5%				

Table 16: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 0.0

wavefront coherence time		Observations		
ms		tau < 3.4	3.4 < tau < 5.8	tau > 5.8
Model data	tau < 3.4	4447	248	516
	3.4 < tau < 5.8	230	3990	516
	tau > 5.8	108	988	5506
PC=87.0%; EBD=3.9%; POD1=92.9%; POD2=76.3%; POD3=84.2%				

Table 17: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

wavefront coherence time		Observations		
ms		tau < 3.4	3.4 < tau < 5.8	tau > 5.8
Model data	tau < 3.4	4780	0	6
	3.4 < tau < 5.8	5	5218	6
	tau > 5.8	0	8	6016
PC=99.9%; EBD=0.0%; POD1=99.9%; POD2=99.8%; POD3=99.8%				

Table 18: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 1.22

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2657	2668	1844
	0.6 < glf < 0.7	1812	2568	2442
	glf > 0.7	511	723	1294
PC=39.5%; EBD=14.3%; POD1=53.4%; POD2=43.1%; POD3=23.2%				

Table 19: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	4345	844	5
	0.6 < glf < 0.7	627	4357	819
	glf > 0.7	8	758	4756
PC=81.5%; EBD=0.1%; POD1=87.2%; POD2=73.1%; POD3=85.2%				

Table 20: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	4385	682	1994
	0.6 < glf < 0.7	84	5244	1632
	glf > 0.7	511	33	3586
PC=80.0%; EBD=15.2%; POD1=88.1%; POD2=88.0%; POD3=49.7%				

Table 21: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	4969	12	7
	0.6 < glf < 0.7	3	5945	7
	glf > 0.7	8	2	5573
PC=99.8%; EBD=0.1%; POD1=99.8%; POD2=99.8%; POD3=99.7%				

Table 22: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

2 DAY1: Contingency tables

Tables in section 2 are included in the range: 23-32.

wind speed ms^{-1}		Observations ws < 4.6	4.6 < ws < 7.4	ws > 7.4
	ws < 4.6	3572	925	148
Model data	4.6 < ws < 7.4	3294	4006	1500
	ws > 7.4	253	2297	7649
PC=64.4%; EBD=1.7%; POD1=50.2%; POD2=55.4%; POD3=82.3%				

Table 23: day1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}		Observations ws < 4.6	4.6 < ws < 7.4	ws > 7.4
	ws < 4.6	6497	454	0
Model data	4.6 < ws < 7.4	622	6370	416
	ws > 7.4	0	404	8881
PC=92.0%; EBD=0.0%; POD1=91.3%; POD2=88.1%; POD3=95.5%				

Table 24: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed ms^{-1}		Observations ws < 12.0	12.0 < ws < 18.0	ws > 18.0
	ws < 12.0	19384	253	0
Model data	12.0 < ws < 18.0	186	2812	115
	ws > 18.0	0	91	803
PC=97.3%; EBD=0.0%; POD1=99.0%; POD2=89.1%; POD3=87.5%				

Table 25: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0) ms^{-1}

wind direction degree		Observations North	East	South	West
Model data	North	9110	594	13	205
	East	97	381	76	0
	South	46	43	2987	840
	West	2930	51	248	4398
PC=76.6%; EBD=0.5%; POD1=74.8%; POD2=35.6%; POD3=89.9%; POD4=80.8%					

Table 26: day1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		Observations North	East	South	West
Model data	North	11775	69	0	231
	East	91	987	10	0
	South	0	13	3238	45
	West	317	0	76	5167
PC=96.1%; EBD=0.0%; POD1=96.7%; POD2=92.3%; POD3=97.4%; POD4=94.9%					

Table 27: day1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		Observations rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	4440	1226	4
	8.9 < rh < 17.9	3290	6028	1264
	rh > 17.9	231	1641	5457
PC=67.5%; EBD=1.0%; POD1=55.8%; POD2=67.8%; POD3=81.1%				

Table 28: day1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		Observations rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	7564	308	0
	8.9 < rh < 17.9	397	8404	179
	rh > 17.9	0	183	6546
PC=95.5%; EBD=0.0%; POD1=95.0%; POD2=94.5%; POD3=97.3%				

Table 29: day1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		Observations rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	22946	13	0
	50.0 < rh < 70.0	25	365	18
	rh > 70.0	0	7	207
PC=99.7%; EBD=0.0%; POD1=99.9%; POD2=94.8%; POD3=92.0%				

Table 30: day1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor		Observations		
mm		pwv < 2.1	$2.1 <$ pwv < 3.7	pwv > 3.7
	pwv < 2.1	5813	0	787
Model data	$2.1 <$ pwv < 3.7	0	0	0
	pwv > 3.7	1240	0	14583
PC=91.0%; EBD=9.0%; POD1=82.4%; POD2=NaN%; POD3=94.9%				

Table 31: day1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor		Observations		
mm		pwv < 2.1	$2.1 <$ pwv < 3.7	pwv > 3.7
	pwv < 2.1	6824	200	0
Model data	$2.1 <$ pwv < 3.7	229	5964	154
	pwv > 3.7	0	61	8991
PC=97.1%; EBD=0.0%; POD1=96.8%; POD2=95.8%; POD3=98.3%				

Table 32: day1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)